Customer Segmentation & Retention Analytics

Power BI Portfolio Project Summary

Tools	Power BI Desktop, Power Query, DAX, Excel/CSV, GitHub
Dataset	Synthetic customer-level snapshot containing 150 customer records
Report	Two-page interactive dashboard for customer segmentation and retention indicators

Project Objective

Build an interactive business intelligence dashboard to explore customer value, purchase behavior, RFM segments, and retention-status indicators. The goal is to help stakeholders identify valuable customer groups and inform targeted marketing and customer engagement decisions.

Data Preparation & Modeling

- Imported the customer dataset into Power BI and prepared it using Power Query.
- Reviewed field names and data types, including join date and last purchase date.
- Created a customer table named DimCustomer and a calendar table named DimDate.
- Related DimDate[Date] to DimCustomer[Join Date] for customer join-month analysis.

DAX & Analytical Measures

- Customer KPIs: Total Customers, Total Historical Spend, Total Recorded Orders, and Average Spend per Customer.
- Repeat behavior: Repeat Customers and Repeat Customer Rate.
- Retention indicators: Retained Customers, At-Risk Customers, Churned Customers, and Retention Rate Proxy.
- RFM analysis: Recency, Frequency, and Monetary scores, total RFM score, and customer segment classification.

Dashboard Features

- KPI cards summarizing customer count, historical spend, repeat customer rate, and at-risk customers.
- Bar chart comparing customer counts by RFM segment.
- Column chart comparing historical spend by RFM segment.
- Donut chart showing calculated retention-status distribution.
- Customer trend by join month and a segment performance matrix.
- Interactive slicers for location, RFM segment, purchase channel, loyalty tier, and gender, plus a date-range slicer.

Business Value

- Compare customer segment size and historical spending to identify high-value groups.
- Monitor customers classified as retained, at risk, or churned using the defined recency rules.
- Prioritize targeted win-back campaigns for at-risk, high-value customers.
- Use loyalty initiatives and personalized offers to support engagement among valuable customer groups.

Key Interpretation & Limitations

The dataset is a customer-level snapshot with one row per customer. Historical spend is not a monthly revenue series. Retention status is a recency-based proxy, and the join-month trend represents customer acquisition timing rather than true monthly retention. Genuine cohort retention and transaction-level revenue trends require transaction history or periodic customer snapshots. The dataset is synthetic and intended for portfolio demonstration.

Project Deliverables

- Power BI report (.pbix)
- Source dataset (.csv or .xlsx)
- Dashboard screenshots for both report pages
- README.md and project documentation
- This project summary PDF

Portfolio positioning
Demonstrates practical skills in data preparation, data modeling, DAX calculations, RFM-based customer segmentation, interactive dashboard design, and business insight communication.